
The Bellman Residual Is Not the Improvement Signal

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Abstract

Actor-critic methods typically assess critic quality using the Bellman residual ϵ_u computed on rollout states. We argue that this quantity is misaligned with policy improvement: it is measured on the wrong distribution and at the wrong scale. We establish an exact identity, $J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k$, where $\hat{\Delta}_k := J(\pi_k) - \mathbb{E}_{s \sim \nu}[V_{k+1}(s)]$ measures the start-state bias of the post-update critic. The Bellman residual does not appear in the improvement decomposition. We further prove a structural limitation: any fixed absolute tolerance on ϵ_u becomes uninformative as π_k converges and $A_k \rightarrow 0$, rendering it unable to certify progress regardless of magnitude.

Empirically, across $\sim 90,000$ updates spanning four configurations (PPO, PPO-MSE, SAC, TRPO) on up to eight environments (263 seeds total), we find that $-\hat{\epsilon}_u$ is not predictive of harmful updates (pooled median AUROC 0.47), while $-\hat{\Delta}_k$ achieves stronger predictive performance (0.68) and dominates $-\hat{\epsilon}_u$ on 256 of 262 paired seeds ($p < 10^{-43}$). We also observe that the surrogate advantage A_k optimized by PPO can become anti-predictive in complex environments (pooled median AUROC 0.39), consistent with overfitting effects that inflate A_k when start-state bias is large. Finally, an additive ν -weighted critic-loss intervention (PPO-NU) fails to make $\hat{\Delta}_k$ prospective, suggesting that the mismatch is not closeable by simple reweighting of a rollout-fit loss. These results suggest that actor-critic monitoring and critic-loss design should move beyond rollout-distribution residuals toward start-state-aware quantities.

1 Introduction

Actor-critic algorithms in deep reinforcement learning (RL)—PPO [Schulman et al., 2017], SAC [Haarnoja et al., 2018], TRPO [Schulman et al., 2015a]—update a policy iteratively against a learned value function. The critic is fit, monitored, and *judged* by its mean squared TD error on the rollout distribution d^{π_k} ; this is the loss every implementation we surveyed minimizes [Huang et al., 2022, Andrychowicz et al., 2021], and the scalar practitioners watch to decide whether the critic is “good enough” to support policy improvement.

The structural problem. Policy improvement, however, is not governed by critic error on d^{π_k} . The expected return $J(\pi) = \mathbb{E}_{s_0 \sim \nu}[V^\pi(s_0)]$ involves the critic only through its accuracy at the start-state distribution ν . The two distributions are not the same: d^{π_k} is the discounted state-visitation distribution from ν , and the difference is non-trivial even on-policy. Actor-critic methods therefore optimize and certify a critic on the wrong distribution.

The consequence is not merely that ϵ_u is a noisy proxy, but that it is often at the wrong scale. As the policy converges and $A_k \rightarrow 0$, any fixed absolute ϵ_u tolerance eventually encodes a return scale

larger than the remaining improvement signal, precluding prospective certification. The work below makes this consequence exact, then confirms it empirically at scale.

Distinction from prior treatments. While classical on-policy improvement bounds treat the d^π/ν mismatch as an approximation gap closed by importance-ratio reweighting or trust-region constraints, we show it is a structural fact. The identity below eliminates ϵ_u entirely at the population level, exposing the residual scaffolding of prior bounds as unnecessary, and fixing a new target for future critic-loss design.

The identity that eliminates the Bellman residual. The mismatch admits an exact algebraic resolution. For any value function V_{k+1} ,

$$J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k, \quad \hat{\Delta}_k := J(\pi_k) - \mathbb{E}_{s \sim \nu}[V_{k+1}(s)]. \quad (1)$$

This is not a bound and not an explanation. It is the ground-truth decomposition of policy improvement, and it reveals why the standard monitor ϵ_u is fundamentally misaligned with the objective: the Bellman residual does not appear in the improvement decomposition at all. While classical bounds introduce ϵ_u as scaffolding, the exact algebraic resolution shows the start-state bias $\hat{\Delta}_k$ is the operative quantity. Furthermore, as we show in Section 5, ϵ_u is also at the wrong *scale*: the relevant tolerance is ϵ_u/A_k , so any fixed absolute tolerance eventually fails to certify even genuine improvement as $A_k \rightarrow 0$.

Empirical confirmation across $\sim 90,000$ updates. The structural prediction is unambiguous: ϵ_u should not track policy improvement, and $\hat{\Delta}_k$ should. We audit 263 seeds across PPO, the PPO-MSE ablation, SAC, and TRPO on up to eight environments. Because ϵ_u measures critic error correctly, but on the wrong distribution, it is *anti-predictive* of policy harm (pooled median AUROC 0.47, below chance — informative but with the wrong sign). Because $\hat{\Delta}_k$ reflects the actual covariate shift, it pools to 0.68 and dominates ϵ_u on **256** of **262** paired seeds (paired Wilcoxon $p < 10^{-43}$). The dominance is uniform across all four configurations, robust across a 170-run hyperparameter factorial, and survives on fully independent held-out rollouts (AUROC 0.61). This mismatch leads to an “ A_k paradox” where the surrogate advantage A_k optimized by PPO can become anti-predictive in complex environments (pooled median AUROC 0.39).

The mismatch is not closeable by reweighting. We further show that adding an explicit start-state penalty to the critic loss fails to make $\hat{\Delta}_k$ prospective, suggesting this mismatch is not a deficiency of MSE, but an intrinsic limitation of the actor–critic optimization order (see Section 7).

2 Related work

Policy-improvement bounds and the on-policy gap. Classical on-policy improvement bounds — CPI [Kakade and Langford, 2002], RPI [Piotto et al., 2013], TRPO [Schulman et al., 2015a], and CPO [Achiam et al., 2017] — confront the same d^π/ν mismatch we identify. Each handles it by a change-of-measure argument: importance-ratio reweighting of the surrogate advantage (CPI) or a trust-region/ D_{KL} constraint that keeps consecutive rollout distributions close (TRPO, CPO). The mismatch survives in the form of a residual scaffolding term proportional to ϵ_u scaled by an unknown density ratio. The identity in Proposition 1 shows this scaffolding is unnecessary at the population level: the gap admits an exact algebraic resolution that eliminates ϵ_u and isolates the start-state bias as the operative quantity. The complementary off-policy distribution-shift and offline RL literature [Islam et al., 2019, ?, ?, Prudencio et al., 2023] addresses a related but distinct mismatch (behavior vs. target policy) often via stationary distribution corrections; however, the gap we identify persists even strictly on-policy.

Critic failure in deep RL. Empirically, Ilyas et al. [2020] established that deep RL critics often fail to fit the true value function even on-policy, observing that the network often solves the supervised task (minimizing TD error) without solving the estimation problem (predicting returns). Andrychowicz et al. [2021] further confirmed that value-loss is poorly correlated with agent performance across a large-scale PPO study. Our identity (Proposition 1) provides the structural explanation for these

empirical findings: because value-loss is minimized over the wrong distribution (d^π), it is dimensionally and structurally decoupled from the start-state error ($\hat{\Delta}_k$) that actually governs the improvement signal.

3 The central lens: an exact identity at the start-state distribution

We consider an infinite-horizon discounted MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \nu)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, reward R , discount $\gamma \in [0, 1)$, and initial-state distribution $\nu \in \Delta(\mathcal{S})$. For a policy π , write V^π for its value function, $J(\pi) = \mathbb{E}_{s_0 \sim \nu}[V^\pi(s_0)]$ for its expected return, T^π for the Bellman operator, and d^π for the discounted state-visitation distribution from ν :

$$(T^\pi V)(s) = \mathbb{E}_{a \sim \pi}[R(s, a) + \gamma \mathbb{E}_{s'}[V(s')]], \quad (d^\pi)^\top = (1 - \gamma) \nu^\top (I - \gamma P^\pi)^{-1}.$$

Consider an iterative algorithm that, at iteration k , holds π_k and a critic estimate V_k , and produces (π_{k+1}, V_{k+1}) . The estimate V_{k+1} need not coincide with $V^{\pi_{k+1}}$.

Proposition 1 (Improvement identity at the initial-state distribution). *For any policies π_k, π_{k+1} and any value function $V_{k+1} : \mathcal{S} \rightarrow \mathbb{R}$,*

$$J(\pi_{k+1}) - J(\pi_k) = A_k - \hat{\Delta}_k, \quad (2)$$

where

$$A_k := \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi_{k+1}}} [(T^{\pi_{k+1}} V_{k+1})(s) - V_{k+1}(s)] \text{ and } \hat{\Delta}_k := J(\pi_k) - \mathbb{E}_{s \sim \nu}[V_{k+1}(s)]. \quad (3)$$

Proof. The Bellman residual at π_{k+1} on V_{k+1} is $T^{\pi_{k+1}} V_{k+1} - V_{k+1} = r^{\pi_{k+1}} + \gamma P^{\pi_{k+1}} V_{k+1} - V_{k+1}$. Substituting $(d^{\pi_{k+1}})^\top = (1 - \gamma) \nu^\top (I - \gamma P^{\pi_{k+1}})^{-1}$ gives

$$A_k = \nu^\top (I - \gamma P^{\pi_{k+1}})^{-1} (r^{\pi_{k+1}} - (I - \gamma P^{\pi_{k+1}}) V_{k+1}) = J(\pi_{k+1}) - \mathbb{E}_\nu V_{k+1}.$$

Subtracting $\hat{\Delta}_k = J(\pi_k) - \mathbb{E}_\nu V_{k+1}$ gives $A_k - \hat{\Delta}_k = J(\pi_{k+1}) - J(\pi_k)$. $\square$

The identity is algebraically equivalent to the Performance Difference Lemma [Kakade and Langford, 2002] with V_{k+1} substituted for V^{π_k} and rearranged to make $\hat{\Delta}_k$ explicit. The insight is not the rearrangement but its consequence: ϵ_u is eliminated from the improvement signal at the population level. Why has ϵ_u persisted as the standard diagnostic despite PDL being textbook? We find that ϵ_u and $\hat{\Delta}_k$ are sufficiently correlated during high-entropy early training (median Spearman $\rho = 0.52$ on Humanoid-v5) that the monitor failure is masked while improvement is large. The mismatch manifests only near convergence — exactly when monitoring matters most.

The identity is exact for any V_{k+1} — we verify it holds to machine precision in tabular RandomMDPs (Section B, median residual 5×10^{-16}). It is not a bound, not a forecasting claim, and contains no Bellman-residual term. The classical CPI/TRPO bound [Kakade and Langford, 2002, Schulman et al., 2015a] introduces ϵ_u only as scaffolding for a change-of-measure step from d^{π_k} to ν ; once the conversion is done exactly, ϵ_u vanishes and the start-state bias $\hat{\Delta}_k$ is what remains. The price is that $\hat{\Delta}_k$ depends on the post-update critic and is retrospective; the gain is that nothing about critic accuracy on d^{π_k} enters at all. Throughout the rest of the paper we treat $\hat{\Delta}_k$ — not the full identity remainder — as the diagnostic of interest, since the empirical work below shows the predictive signal lives almost entirely in this term.

Practical estimators. In a deep RL implementation, the population quantities of Equation (3) are not directly observable. Note that while the exact population identity (Equation (2)) requires the post-update critic V_{k+1} , standard deep RL implementations compute the surrogate $\hat{A}_k$ via GAE using the pre-update critic V_k . We evaluate the standard estimator to reflect the actual optimization signal utilized in practice. The estimators we use are:

$$\hat{A}_k := \mathbb{E}_{(s,a) \sim B_k} \left[\frac{\pi_{k+1}(a|s)}{\pi_k(a|s)} \hat{A}_{V_k}^{\text{GAE}}(s, a) \right], \quad (4)$$

$$\hat{\Delta}_k := \hat{J}(\pi_k) - \frac{1}{N_{\text{eps}}} \sum_{i=1}^{N_{\text{eps}}} V_{k+1}(s_0^{(i)}), \quad (5)$$

$$\hat{\Delta}_k^{\text{res}} := \frac{1}{1-\gamma} \mathbb{E}_{(s,a,s') \sim B_k} [r(s, a) + \gamma V_{k+1}(s') - V_{k+1}(s)]. \quad (6)$$

Here B_k is the on-policy rollout batch under π_k , $\hat{A}^{\text{GAE}}$ is the GAE [Schulman et al., 2015b] advantage at the pre-update critic V_k , $\hat{J}(\pi_k)$ is the mean episode return on the rollout, and N_{eps} is the number of episodes sampled in the rollout, with $s_0^{(i)}$ being the start state of the i -th episode. We adopt the start-state form Equation (5) for $\hat{\Delta}_k$ throughout; the alternative residual-based estimator Equation (6) has lower per-update variance but agrees only in expectation (Section D). The estimator $\hat{\Delta}_k$ is biased relative to the population Δ_k because $\hat{J}(\pi_k)$ has finite-sample variance, and the rollout-conditional start-state mean diverges from the true unconditional expectation $\mathbb{E}_\nu$.

4 Audit setup

We audit four actor–critic configurations spanning three algorithm families.

Algorithms and environments. PPO [Schulman et al., 2017] on the canonical 8-environment suite at 200k–500k timesteps each: discrete-action LunarLander-v3, CartPole-v1, Acrobot-v1, and MuJoCo continuous control on MuJoCo [Todorov et al., 2012] Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, Humanoid-v5; 20 seeds per environment (160 runs). PPO-MSE: an ablation that replaces PPO’s clipped value loss with a pure mean-squared TD error, on the same 8 envs $\times$ 5 seeds. SAC [Haarnoja et al., 2018] on five MuJoCo environments (Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, Humanoid-v5), 6–10 seeds each (46 runs). TRPO [Schulman et al., 2015a] on LunarLander-v3 and Hopper-v5 (10 seeds each; 1 LunarLander seed stalled yielding 9 valid seeds). PPO-NU: an ablation adding an explicit ν -weighted MSE loss term, on Hopper-v5 and Humanoid-v5 (15 seeds total). Seeds stalled below 10 updates are excluded from per-seed AUROC; 263 seeds remain (Table 1, n column). Total: $\sim 90,000$ actor–critic updates with paired (diagnostic, ground-truth ΔJ_k) records.

For each update we log $\hat{A}_k$, $\hat{\Delta}_k$ (the start-state estimator, eq. (5)), $\hat{\epsilon}_u$ (post-critic-update mean squared TD error on the rollout — the canonical scalar minimized in PPO/SAC/TRPO codebases), and the ground-truth return $\hat{J}(\pi_k)$. The harmful label is $1[\hat{\Delta J}_k < 0]$, where $\hat{\Delta J}_k = \hat{J}(\pi_{k+1}) - \hat{J}(\pi_k)$ from successive rollout batches. The harmful label is undefined on the final update of each run.

Hyperparameter factorials. To rule out tuning artifacts we run three factorial sweeps. LunarLander: 4×4 over value-epochs $\in \{5, 10, 20, 40\}$ and clip- $\epsilon \in \{0.1, 0.2, 0.3, 0.4\}$, 5 seeds per cell (80 runs). HalfCheetah and Hopper: 3×2 over value-epochs $\in \{5, 10, 20\} \times$ clip- $\epsilon \in \{0.1, 0.2\}$ plus a 3-cell value-learning-rate sweep at the default cell, 5 seeds each (45 runs per env). All factorials use the same logging schema. Full details in Section J.

5 ϵ_u is the wrong scale: a structural incompatibility

The identity already shows ϵ_u is the wrong *quantity*. We now show it is also the wrong *scale* — a separate, complementary failure rooted in the same mismatch. Consider the mixture-policy update used in CPI [Kakade and Langford, 2002] and TRPO [Schulman et al., 2015a]: $\pi_{k+1}^\alpha = \alpha \bar{\pi}_k + (1 - \alpha) \pi_k$ with $\bar{\pi}_k$ greedy on V_{k+1} and $\alpha \in [0, 1]$. The CPI/TRPO improvement bound is a concave quadratic

$$\Psi(\alpha) = c_1 \alpha A_k - \alpha^2 \frac{c_2 C_k^2}{1 - \gamma} - \alpha \frac{c_3 C_k \epsilon_u}{1 - \gamma},$$

where $C_k = \|\bar{\pi}_k - \pi_k\|_{1, d^{\pi_k}}$ is the policy-change norm. Since Ψ is a downward parabola with $\Psi(0) = 0$, its unconstrained maximum on $\alpha > 0$ is positive iff $\Psi'(0) > 0$, i.e. $2(1 - \gamma)A_k > C_k \epsilon_u$. Equivalently, the bound certifies progress at some step size if and only if the dimensionless ratio

$$\tilde{\rho}_k := \frac{2(1 - \gamma)A_k}{C_k \epsilon_u}$$

exceeds one. The certifying object is ϵ_u/A_k , not ϵ_u . As $\pi_k \rightarrow \pi^*$, $A_k \rightarrow 0$; the residual need not vanish in tandem when the critic is misspecified or undertrained. Any threshold “critic loss is small enough,” fixed in absolute terms, eventually fails to certify even genuine improvement.

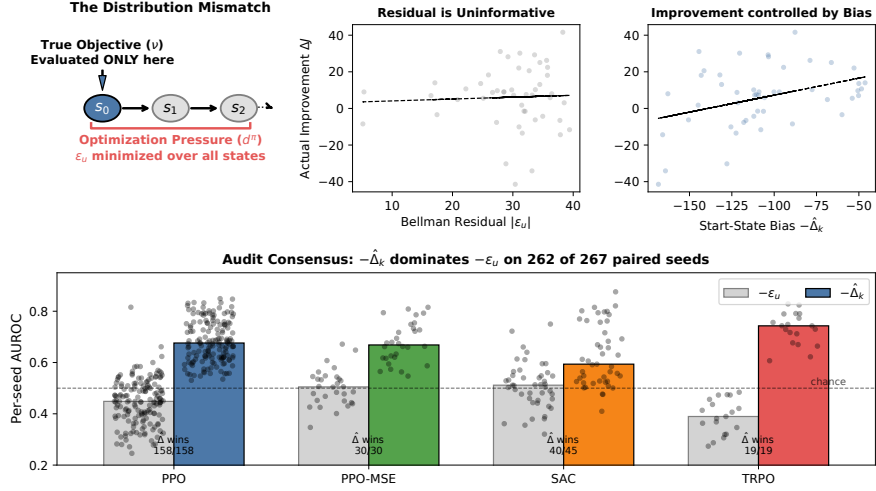


Figure 1: **The distribution mismatch decouples value-loss from improvement.** (Panel A) The Bellman residual ϵ_u is minimized over the rollout distribution d^π (all states traversed in an episode). However, the true objective—policy improvement—depends exclusively on critic accuracy at the start states ν . (Panel B) Because it measures the wrong distribution, the absolute Bellman residual $|\epsilon_u|$ is near-blind to ground-truth improvement ΔJ_k (Humanoid-v5, representative run). (Panel C) In contrast, measuring the critic’s error exactly at the start states (the start-state bias $-\hat{\Delta}_k$) exposes the operative relationship. (Panel D) Median per-seed AUROC for harm prediction: $-\epsilon_u$ (gray) sits below chance across PPO and TRPO and at chance for SAC; $-\hat{\Delta}_k$ (colored) is reliably predictive in every algorithm. Full per-environment table is Table 1.

Proposition 2 (Fixed absolute tolerances are structurally incompatible). *Fix any $\gamma \in (0, 1)$ and any $\epsilon^{\text{fixed}} > 0$. Let M_0 be any MDP with bounded rewards in which there exists a policy sequence (π_k) with $A_k(M_0) > 0$ along the sequence. Then there is a single linear rescaling of the rewards of M_0 , by a constant $c \propto \epsilon^{\text{fixed}}$, producing an MDP M that admits a critic sequence (V_{k+1}) satisfying the Relative Policy Improvement invariant $T^{\pi_k} V_{k+1} \geq V_{k+1}$ [Pirrotta et al., 2013] with $\epsilon_{u,k} = \epsilon^{\text{fixed}}$, for $\epsilon_{u,k}$ evaluated in any positively-homogeneous norm, yet for which $\hat{\rho}_k \leq 1 - \gamma < 1$ for all k .*

The proof (Section C) is a *scale-invariance argument*, not a pathological construction. The dimensionless ratio $\hat{\rho}_k$ is invariant under any common scaling of rewards, but the absolute tolerance ϵ^{fixed} is not. Consequently the proof works on *any* non-trivial MDP, transformed by a single linear rescaling chosen to make the policy-improvement signal small relative to the absolute tolerance. The dimensional content is direct: ϵ_u and A_k both have units of return-magnitude, so a fixed absolute tolerance implicitly encodes a return-scale, and any MDP whose rewards live below that scale evades certification — regardless of how small ϵ^{fixed} is. Equivalently, since $A_k \rightarrow 0$ as $\pi_k \rightarrow \pi^*$ in any MDP, the regime in which the absolute tolerance is too coarse is reached either by reward rescaling or by approaching a local optimum — both entirely generic. “Make ϵ_u smaller” is not a recovery strategy; the mismatch is dimensional, not numerical. This is the theoretical counterpart to the empirical observation of Section 6: the absolute Bellman residual is uncorrelated with the direction of the next policy update because it cannot be on the right scale to begin with.

6 Results

The mismatch makes a sharp prediction: a scalar minimized on d^{π_k} should not track improvement on ν , while a scalar measured on ν should. Figure 1 previews the pattern; Table 1 reports the full per-environment median AUROC for $-\hat{\epsilon}_u$ and $-\hat{\Delta}_k$ across all four configurations. Both predictions hold uniformly. Because $\hat{\epsilon}_u$ measures critic error on the wrong distribution, it pools to AUROC 0.47 across 263 seeds — below chance, informative but with the wrong sign: strictly anti-predictive on 6

Table 1: **Per-environment median harm-prediction AUROC, by algorithm.** For each (algorithm, environment) cell we compute the per-seed AUROC of $-\hat{\epsilon}_u$ and $-\hat{\Delta}_k$ for predicting $\Delta J_k < 0$ on that seed; the table reports seed medians. ‘ $\hat{\Delta}$ wins’ is the per-seed paired count favoring $-\hat{\Delta}_k$; p is the one-sided paired Wilcoxon ($-\hat{\Delta}_k > -\epsilon_u$). ‘base’ is the harmful-update fraction (chance AUROC is 0.5; chance AUPRC equals the base rate). Cells with $n < 5$ seeds report ‘—’ for the Wilcoxon p .

Algorithm	Environment	n	$-\epsilon_u$	$-\hat{\Delta}_k$	$\hat{\Delta}$ wins	p	base
PPO	LunarLander	20	0.45	0.72	20/20	1e-06	0.48
PPO	CartPole	19	0.38	0.61	19/19	2e-06	0.29
PPO	Acrobot	20	0.32	0.81	19/19	2e-06	0.42
PPO	Hopper	20	0.46	0.66	20/20	1e-06	0.31
PPO	HalfCheetah	20	0.55	0.63	20/20	1e-06	0.44
PPO	Walker2d	20	0.45	0.62	20/20	1e-06	0.41
PPO	Ant	20	0.43	0.67	20/20	1e-06	0.48
PPO	Humanoid	20	0.52	0.75	20/20	1e-06	0.46
PPO-MSE	LunarLander	5	0.50	0.72	5/5	0.031	0.48
PPO-MSE	CartPole	5	0.48	0.61	4/5	0.062	0.28
PPO-MSE	Acrobot	5	0.27	0.80	5/5	0.031	0.47
PPO-MSE	Hopper	5	0.51	0.73	5/5	0.031	0.31
PPO-MSE	HalfCheetah	5	0.58	0.63	5/5	0.031	0.44
PPO-MSE	Walker2d	5	0.43	0.62	5/5	0.031	0.41
PPO-MSE	Ant	5	0.45	0.59	5/5	0.031	0.49
PPO-MSE	Humanoid	5	0.53	0.75	5/5	0.031	0.47
PPO-NU	Hopper	5	0.39	0.65	5/5	0.031	0.33
PPO-NU	HalfCheetah	5	0.54	0.58	5/5	0.031	0.42
PPO-NU	Humanoid	5	0.52	0.76	5/5	0.031	0.47
SAC	Hopper	9	0.56	0.72	8/9	0.027	0.40
SAC	HalfCheetah	10	0.51	0.56	9/10	0.003	0.36
SAC	Walker2d	10	0.54	0.78	10/10	0.001	0.45
SAC	Ant	10	0.47	0.54	8/10	0.019	0.51
SAC	Humanoid	6	0.49	0.52	5/6	0.156	0.51
TRPO	LunarLander	9	0.39	0.79	9/9	0.002	0.44
TRPO	Hopper	10	0.39	0.71	10/10	0.001	0.33
<i>pooled</i>	<i>PPO</i>	159	0.45	0.68	158/158	6e-28	—
<i>pooled</i>	<i>PPO-MSE</i>	40	0.48	0.67	39/40	2e-12	—
<i>pooled</i>	<i>PPO-NU</i>	15	0.52	0.65	15/15	3e-05	—
<i>pooled</i>	<i>SAC</i>	45	0.51	0.59	40/45	1e-08	—
<i>pooled</i>	<i>TRPO</i>	19	0.39	0.74	19/19	2e-06	—
ALL	(4 algos, 23 cells)	263	0.46	0.67	256/262	4e-44	—

of 8 PPO environments (Acrobot 0.32, CartPole 0.38, Ant 0.43, Hopper, Walker2d, LunarLander all ≤ 0.46) and on both TRPO environments (0.39 each). Because $-\hat{\Delta}_k$ measures error on the operative distribution ν , it pools to 0.68 and beats $\hat{\epsilon}_u$ on 256 of 262 paired seeds (paired Wilcoxon $p < 10^{-43}$). Every per-algorithm slice shows the same direction: PPO (158/158, $p < 10^{-27}$), the PPO-MSE ablation (39/40, $p < 10^{-12}$), SAC (40/45, $p < 10^{-8}$), and TRPO (19/19, $p < 10^{-6}$). While the direction is uniform, we note that the SAC signal is generally weaker than PPO (pooled 0.59 vs 0.68), and the Humanoid SAC results (5 seeds) are likely underpowered to achieve statistical significance on a per-environment basis.

The PPO-MSE ablation: ϵ_u fails on the rollout distribution. A reasonable objection is that PPO’s value loss is *clipped*, so $\hat{\epsilon}_u$ might fail because it is the wrong scalarization rather than because it is on the wrong distribution. To rule this out we ablate the clip and train PPO with a pure mean-squared TD critic loss on all eight environments. Table 1 (PPO-MSE rows) shows the same pattern: $-\hat{\epsilon}_u$

Table 2: **Per-environment A_k AUROC and Spearman correlation with $\hat{\Delta}_k$ on PPO updates.** Positive Spearman ρ indicates A_k and $\hat{\Delta}_k$ co-rise. The “ A_k paradox” (counter-predictive A_k AUROC < 0.5) holds where this coupling is positive (Acrobot, Humanoid, HalfCheetah, Ant). The dominance of $-\hat{\Delta}_k$ over $-\epsilon_u$ in Table 1 holds regardless.

Env	Acrobot	CartPole	Humanoid	HalfCheetah	Ant	LunarLander	Walker2d	Hopper
A_k AUROC	0.26	0.53	0.33	0.39	0.39	0.35	0.48	0.56
$\rho(A_k, \hat{\Delta}_k)$	+0.68	−0.16	+0.57	+0.66	+0.49	+0.22	−0.23	−0.42
n updates	2,920	1,847	4,684	2,920	2,920	2,920	2,920	2,822

at chance (0.48 pooled), $-\hat{\Delta}_k$ at 0.67. The failure is not a clipping artefact and not a scalarization choice (Section E shows the same with sup-norm and 95th-percentile residuals); it is the distribution.

Where the predictive signal lives. Auditing the identity’s two terms separately shows the predictive signal lives almost entirely in $\hat{\Delta}_k$: across the 8 PPO environments the $-\hat{\Delta}_k$ AUROC matches the AUROC of the full identity remainder $A_k - \hat{\Delta}_k$ to within ± 0.01 , while A_k -alone is pooled 0.39 — *below chance*.

6.1 The A_k paradox: the mechanism of optimization-signal corruption

A_k pooling to AUROC 0.39 is a major empirical finding: the quantity PPO maximizes is, in complex environments, evidence *against* improvement. The mechanism of the paradox is visible in the coupling of A_k and $\hat{\Delta}_k$: A_k is largest precisely when $\hat{\Delta}_k$ is large (Figure 2, Panel B). We hypothesize a failure mode where the critic overfits to mid-trajectory states, causing it to badly underestimate start-state values. Because GAE accumulates temporal difference errors, underestimating $V(s_0)$ while accurately estimating mid-trajectory $V(s_t)$ artificially inflates the temporal difference errors early in the rollout. The resulting surrogate A_k is then inflated for actions that merely survive the start state, driving the policy to make an aggressive but uninformed step. One might suspect this is an artifact of sparse start-state sampling, but evaluating the paradox across all eight PPO environments reveals no significant correlation between episodes-per-rollout and A_k AUROC ($\rho = -0.24, p = 0.57, n = 8$ environments), suggesting the paradox is not explained by sampling density alone. The paradox holds in densely-sampled Acrobot (A_k AUROC 0.26) and sparsely-sampled HalfCheetah (0.39) alike; in fact, 6 of the 8 evaluated environments show A_k AUROC below 0.5, with only CartPole (0.53) and Hopper (0.56) avoiding strictly anti-predictive optimization signals.

This confirms the distribution mismatch corrupts the optimization signal itself, not just the monitor.

Comparison to alternative diagnostics. We evaluate $\hat{\Delta}_k$ against standard PPO diagnostics on Humanoid-v5 to confirm its distinctive informativeness. The median AUROC across 20 seeds for $-\hat{\Delta}_k$ is 0.75 (Table 1), while the standard rollout value-loss ϵ_u (median 0.51), a held-out evaluation MSE (median 0.61, evaluated on an independent rollout), policy entropy (median 0.58), and grad norm (median 0.52) all provide substantially weaker predictive signals across the seeds. This suggests the start-state bias is not merely tracking general training stability or convergence rate, but is uniquely sensitive to the distribution mismatch that governs policy improvement.

Heterogeneity and episode density. The $\hat{\Delta}_k$ signal strength (Table 1) varies from 0.56 (HalfCheetah) to 0.81 (Acrobot). This heterogeneity is well-explained by the start-state sampling density: $\hat{\Delta}_k$ AUROC correlates strongly ($\rho = 0.68$) with the number of episodes sampled per rollout. In “episode-sparse” environments like HalfCheetah, the start-state mean $\hat{\Delta}_k$ is a noisier estimator of the population bias, though it remains superior to the rollout-fit ϵ_u regardless.

7 Why the mismatch is not closeable by simple reweighting

The mismatch sets a hard limit on how the diagnostic can be used. $\hat{\Delta}_k$ measures critic error on ν , but it does so using the *post-update* critic V_{k+1} , so it is intrinsically retrospective.

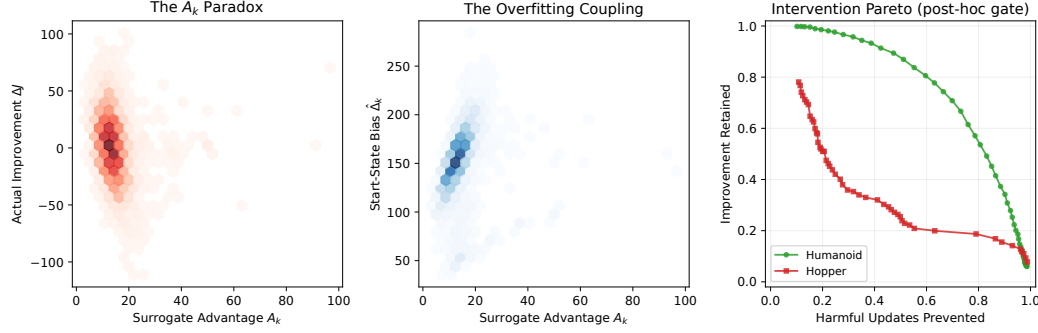


Figure 2: **Mechanism and Intervention.** (Panel A) The A_k paradox: surrogate advantage is anti-predictive of improvement in complex environments. (Panel B) The mechanism: A_k is inflated exactly when start-state bias $\hat{\Delta}_k$ is large, reflecting critic overfitting. (Panel C) Post-hoc gate Pareto frontier: rejecting updates with high $\hat{\Delta}_k$ retains most improvement while preventing the majority of harmful updates.

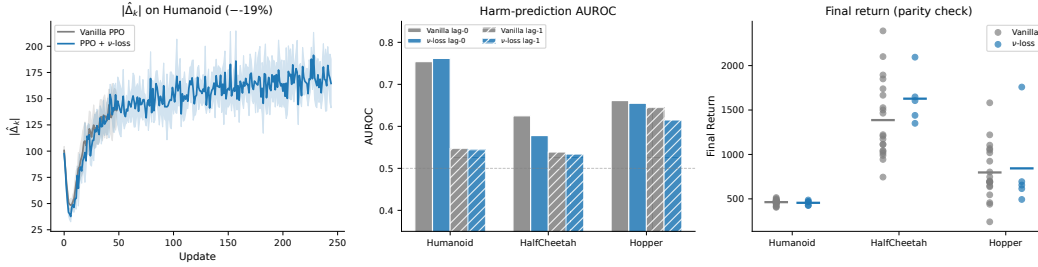


Figure 3: **Targeting the bias directly.** Training with a ν -weighted critic loss (*PPO-NU*) fails to reduce the magnitude of the start-state bias $|\hat{\Delta}_k|$ (Panel A) and leaves the lag-1 prospective AUROC at chance across environments (Panel B), while maintaining final-return parity with vanilla PPO (Panel C).

235 **The lag-1 collapse: the diagnostic does not look one step ahead.** Evaluating the step- k metric
 236 against the step- $(k+1)$ harmful label collapses the AUROC to chance (pooled lag-1 AUROC 0.51).
 237 The signal is concurrent but the post-hoc dependence on V_{k+1} blocks one-step-ahead use.

238 **The mismatch is not closeable by simple additive reweighting (PPO-NU).** The natural fix is
 239 to add the missing distribution to the loss. We test this directly in an ablation we denote *PPO-NU*,
 240 which adds an explicit MSE term at start states s_0 to the standard PPO objective. Figure 3: the
 241 additive penalty does not drive $|\hat{\Delta}_k|$ below vanilla, and the lag-1 prospective AUROC remains at
 242 chance (pooled 0.54 vs vanilla 0.56). Final return is at parity. The diagnostic remains retrospective
 243 even when targeted by the loss.

244 We read this null result as evidence that the mismatch is not closeable by simple additive reweighting.
 245 Adding ν to the critic objective is the most direct attack on the problem, yet it leaves the retrospective
 246 boundary intact. The structural limit is not a deficiency of MSE or clipping but a property of the
 247 actor-critic optimization order: the post-update critic V_{k+1} is fit to a rollout under π_k , not π_{k+1} . We
 248 can formalize what the null ν -loss empirically demonstrates. We provide a formal proof in Section I
 249 that any scalar diagnostic available at step k that relies only on step- k rollout data is fundamentally
 250 decoupled from the next-step improvement signal.

251 **Theorem** (Prospective limit; informal, see Section I for formal statement and proof). *No rollout-only*
 252 *scalar diagnostic can prospectively certify improvement under typical conditions of exploration and*
 253 *finite-sample or function-approximation fallibility.*

254 **Operational relevance: a post-hoc gate.** While standard optimization cannot prospectively fix
 255 the bias, practitioners can still exploit its retrospective nature. We evaluate a simple checkpoint-

and-rollback pattern: compute V_{k+1} on a candidate update, evaluate $\hat{\Delta}_k$, and roll back if $\hat{\Delta}_k > \tau$. Sweeping τ post-hoc on two environments (Figure 2, Panel C) shows the gate carries non-trivial information beyond a single environment: on Humanoid-v5 (canonical high-variance), a threshold flagging the worst updates retains $\sim 80\%$ of improvement while removing $\sim 60\%$ of harm; on MuJoCo [Todorov et al., 2012] Hopper-v5 (episode-sparse, smaller MDP), the same sweep produces a shallower but still informative frontier, retaining $\sim 80\%$ of improvement while removing $\sim 30\%$ of harm. Active gated training on Humanoid-v5 reaches mean final return within 2% of vanilla and reduces per-update return variance by $\sim 11\%$. However, active gating on Hopper at a matched rejection rate over-rejects and collapses learning. This confirms the gate is a diagnostic instrument rather than a universal drop-in safe-RL controller. The active-training collapse on Hopper is consistent with $\hat{\Delta}_k$ being a noisier estimator in episode-sparse environments — we do not claim universal operational gains.

8 Discussion

The Bellman residual is not the quantity that governs policy improvement in actor-critic deep RL. The identity eliminates ϵ_u and isolates the start-state critic bias as the operative scalar; the no-go shows fixed absolute ϵ_u -tolerances are structurally incompatible with the improvement signal; the audit confirms it across 263 seeds and three algorithm families; and the null ν -weighted-loss result establishes that the mismatch is not closeable by additive reweighting of the critic objective. The combined contribution is not a drop-in controller but a structural target: any critic objective that aims to prospectively certify improvement must escape the rollout-only class characterized by Theorem 1.

What this constrains for critic-loss design. Two natural exits from the rollout-only class are (i) implicit objectives that re-evaluate V at π_{k+1} -rollouts in the spirit of Kostrikov et al. [2022], and (ii) Lipschitz-controlled critics where spectral normalization [Miyato et al., 2018, Bjorck et al., 2021] bounds $|\hat{\Delta}_k| \leq L \cdot W_1(\nu, d^{\pi_k})$. We leave their construction to future work and view our contribution as fixing the target the construction must aim at. In the meantime, $\hat{\Delta}_k$ has immediate value as a concurrent diagnostic: in the value-lr sweep (Figure 4, bottom) it separates stable from diverging configurations within the first 10% of training, long before mean return becomes informative.

Limitations. $\hat{\Delta}_k$ inherits the variance of $\hat{J}(\pi_k)$ on environments with few complete episodes per rollout (HalfCheetah ~ 2 , Ant ~ 3); dominance over $\hat{\epsilon}_u$ holds nevertheless. The lag-1 collapse and null ν -loss together establish that $\hat{\Delta}_k$ cannot serve as an online controller; the result is a structural correction, not an algorithm. The SAC and TRPO grids are smaller than PPO’s, so the cross-algorithm conclusion rests on uniform direction across all four configurations rather than per-cell statistical power; the SAC-Humanoid audit ($n = 6$, $p = 0.16$) is the one configuration where dominance is directional rather than significant.

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329 A Hyperparameter factorials

330 The dominance of start-state bias is not specific to a single PPO configuration. Figure 4 reports per-
331 cell concurrent AUROCs across three factorials: a 4×4 value-epochs $\times$ clip- ϵ grid on LunarLander,
332 a 3×2 grid on HalfCheetah, and a 3×2 grid on Hopper, with an additional $3 \times$ value-learning-rate
333 sweep at the default cell of HalfCheetah and Hopper (the configuration that most directly varies critic
334 convergence rate). Across every cell of every environment, $-\hat{\Delta}_k$ concurrent AUROC stays above
335 0.55, while $-\hat{\epsilon}_u$ stays at or below 0.55. The dominance is not driven by a particular learning rate,
336 clip threshold, or critic update count.

337 B Tabular verification of the identity

338 We instantiate REINFORCE on RandomMDPs with $(|S|, |A|, \gamma) \in$
339 $\{(64, 4, 0.95), (128, 8, 0.90), (32, 2, 0.99)\}$. Each iteration uses sampled-rollout estimates of
340 Q for the gradient (2,000 transitions), and a sub-converged TD critic (50 TD steps). We compute A_k
341 via Equation (3) and $\hat{\Delta}_k$ via the trained critic, with ΔJ_k obtained exactly via $\nu^\top (I - \gamma P^\pi)^{-1} r^\pi$.
342 Figure 5: median residual 5×10^{-16} , max 4×10^{-13} . The order-of-magnitude variation across
343 families is the $(1 - \gamma)^{-1}$ amplification expected from the matrix conditioning of $(I - \gamma P^\pi)^{-1}$ at
344 $\gamma = 0.99$.

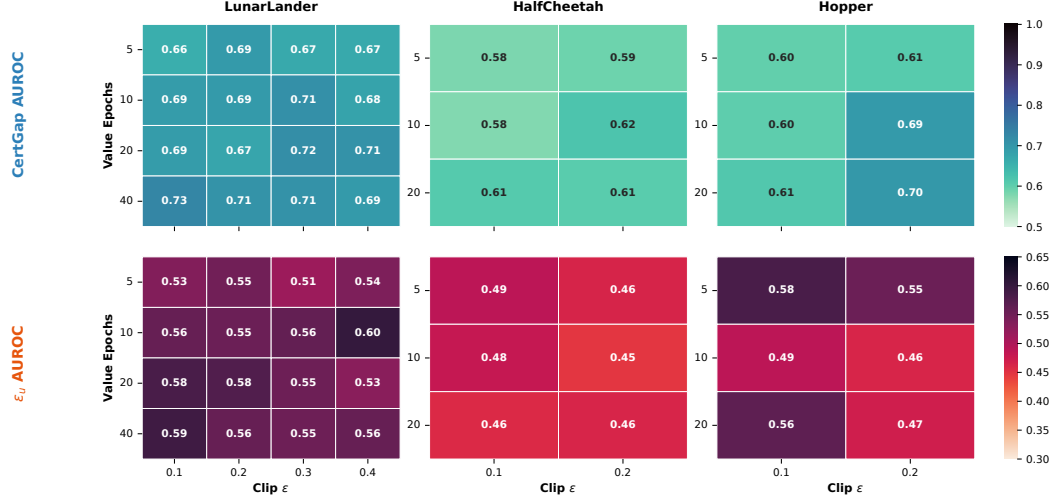


Figure 4: **Hyperparameter robustness** across LunarLander 4×4 , HalfCheetah 3×2 , and Hopper 3×2 factorials, 5 seeds per cell. *Top row*: $-\hat{\Delta}_k$ concurrent AUROC, dark (informative) on every cell. *Bottom row*: $-\hat{\epsilon}_u$ concurrent AUROC, near-chance on every cell. The pattern of the main text is not specific to a single tuning of PPO.

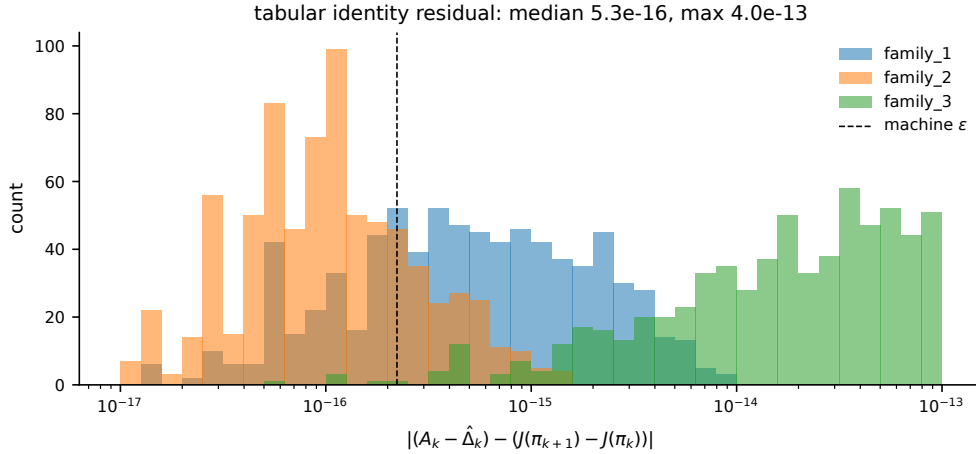


Figure 5: **The improvement identity (Proposition 1) holds to machine precision in tabular RandomMDPs.** Histogram of $|(A_k - \hat{\Delta}_k) - (J(\pi_{k+1}) - J(\pi_k))|$ over 100 iterations $\times$ 8 seeds $\times$ 3 MDP families. Median residual 5×10^{-16} , max 4×10^{-13} . Function approximation is not the cause.

345 C Proof of Proposition 2

346 The proof is a scale-invariance argument: $\tilde{\rho}_k$ is a dimensionless ratio invariant under reward scaling,
 347 while the absolute tolerance ϵ^{fixed} is not. Consequently any fixed ϵ^{fixed} becomes vacuous when applied
 348 to MDPs whose reward magnitude lies below the scale that tolerance encodes.

349 **Setup.** Let $M_0 = (\mathcal{S}, \mathcal{A}, r_0, P, \gamma, \nu)$ be any MDP with bounded rewards $\|r_0\|_\infty \leq R_{\max} < \infty$ in
 350 which there is a policy sequence (π_k) with $A_k(M_0) > 0$ for all k (any policy-gradient trajectory from
 351 a sub-optimal initialization in any non-trivial MDP suffices). Let $A_{\max} := \sup_k \|A_{M_0}^{\pi_k}\|_\infty$. Bounded
 352 rewards give $|A_{M_0}^{\pi_k}(s, a)| \leq 2R_{\max}/(1 - \gamma)$, so $A_{\max} < \infty$. Define $M := (\mathcal{S}, \mathcal{A}, c \cdot r_0, P, \gamma, \nu)$ by
 353 scaling the rewards of M_0 by

$$c := \frac{\epsilon^{\text{fixed}}}{2A_{\max}} \in (0, \infty).$$

Linearity of V^π in the reward signal gives $V_M^\pi = c \cdot V_{M_0}^\pi$, $Q_M^\pi = c \cdot Q_{M_0}^\pi$, $A_M^\pi = c \cdot A_{M_0}^\pi$. The discounted state-visitation distribution d^π depends only on (π, P, ν, γ) , so $d_M^{\pi_k} = d_{M_0}^{\pi_k}$, and hence $C_k(M) = C_k(M_0)$ and $A_k(M) = c \cdot A_k(M_0)$.

Critic sequence. Define

$$V_{k+1} := V_M^{\pi_k} - \frac{\epsilon^{\text{fixed}}}{1-\gamma} \mathbf{1}.$$

For each state s ,

$$\begin{aligned} T_M^{\pi_k} V_{k+1}(s) &= \mathbb{E}_{a \sim \pi_k(\cdot|s), s' \sim P(\cdot|s,a)} \left[r_M(s, a) + \gamma \left(V_M^{\pi_k}(s') - \frac{\epsilon^{\text{fixed}}}{1-\gamma} \right) \right] \\ &= T_M^{\pi_k} V_M^{\pi_k}(s) - \frac{\gamma \epsilon^{\text{fixed}}}{1-\gamma} = V_M^{\pi_k}(s) - \frac{\gamma \epsilon^{\text{fixed}}}{1-\gamma}, \end{aligned}$$

using $T_M^{\pi_k} V_M^{\pi_k} = V_M^{\pi_k}$. Subtracting $V_{k+1}(s) = V_M^{\pi_k}(s) - \epsilon^{\text{fixed}}/(1-\gamma)$,

$$T_M^{\pi_k} V_{k+1}(s) - V_{k+1}(s) = \frac{\epsilon^{\text{fixed}}}{1-\gamma} - \frac{\gamma \epsilon^{\text{fixed}}}{1-\gamma} = \epsilon^{\text{fixed}} \geq 0,$$

so the RPI invariant holds and $\epsilon_{u,k} := \|T_M^{\pi_k} V_{k+1} - V_{k+1}\|_\infty = \epsilon^{\text{fixed}}$.

Surrogate advantage with V_{k+1} . Let $\bar{\pi}_k := \arg \max_a Q_{V_{k+1}}(\cdot, a)$ where $Q_{V_{k+1}}(s, a) := r_M(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} [V_{k+1}(s')]$. Since V_{k+1} differs from $V_M^{\pi_k}$ only by a state-independent constant, $Q_{V_{k+1}}(s, a) = Q_M^{\pi_k}(s, a) - \gamma \epsilon^{\text{fixed}}/(1-\gamma)$, so $\arg \max_a Q_{V_{k+1}}(s, a) = \arg \max_a Q_M^{\pi_k}(s, a)$, i.e., $\bar{\pi}_k$ is the greedy policy with respect to $V_M^{\pi_k}$. Computing the surrogate,

$$\begin{aligned} A_k(M) &= \mathbb{E}_{s \sim d^{\pi_k}} \left[\sum_a (\bar{\pi}_k - \pi_k)(a|s) (Q_{V_{k+1}}(s, a) - V_{k+1}(s)) \right] \\ &= \mathbb{E}_{s \sim d^{\pi_k}} \left[\sum_a (\bar{\pi}_k - \pi_k)(a|s) (A_M^{\pi_k}(s, a) + \epsilon^{\text{fixed}}) \right] \\ &= \mathbb{E}_{s \sim d^{\pi_k}} \left[\sum_a (\bar{\pi}_k - \pi_k)(a|s) A_M^{\pi_k}(s, a) \right] \\ &= c \cdot \mathbb{E}_{s \sim d^{\pi_k}} \left[\sum_a (\bar{\pi}_k - \pi_k)(a|s) A_{M_0}^{\pi_k}(s, a) \right], \end{aligned}$$

By Hölder applied per state, $|\sum_a (\bar{\pi}_k - \pi_k)(a|s) A_{M_0}^{\pi_k}(s, a)| \leq \|\bar{\pi}_k(\cdot|s) - \pi_k(\cdot|s)\|_1 \cdot \|A_{M_0}^{\pi_k}(s, \cdot)\|_\infty$. Taking expectation and using $A_{\max} \geq \|A_{M_0}^{\pi_k}\|_\infty$,

$$A_k(M) \leq c \cdot A_{\max} \cdot C_k.$$

Conclusion. Substituting $c = \epsilon^{\text{fixed}}/(2A_{\max})$ into the definition of $\tilde{\rho}_k$,

$$\tilde{\rho}_k = \frac{2(1-\gamma) A_k(M)}{C_k \cdot \epsilon^{\text{fixed}}} \leq \frac{2(1-\gamma) \cdot c \cdot A_{\max}}{\epsilon^{\text{fixed}}} = 1 - \gamma < 1$$

for all k . □

D Estimator agreement (start-state vs. residual-based $\hat{\Delta}_k$)

The start-state estimator Equation (5) and the alternative residual-based estimator $\hat{\Delta}_k^{\text{res}} = \mathbb{E}_{B_k} [r + \gamma V_{k+1}(s') - V_{k+1}(s)]/(1-\gamma)$ are different empirical proxies for the same population $\hat{\Delta}_k$. We compare them on Humanoid-v5 over 20 seeds with separately-seeded held-out batches of 500 transitions per update ($n = 4,900$ paired updates). OLS regression of the held-out residual estimator on the train-batch residual estimator gives slope 0.97, intercept +2.15, Pearson $r = 0.80$ (95% CI [0.79, 0.81]), $R^2 = 0.64$. The estimators agree in expectation (since $\mathbb{E}_{d^{\pi_k}} [T^{\pi_k} V - V]/(1-\gamma) = J(\pi_k) - \mathbb{E}_\nu V$, which implies slope near 1) but the per-update agreement is moderate ($R^2 = 0.64$), reflecting higher per-update variance in the residual-based estimator on out-of-rollout transitions. We report the start-state form throughout because it is faithful to the formal definition Equation (5); the alternative form has more samples per update but is closer to the rollout that just trained the critic.

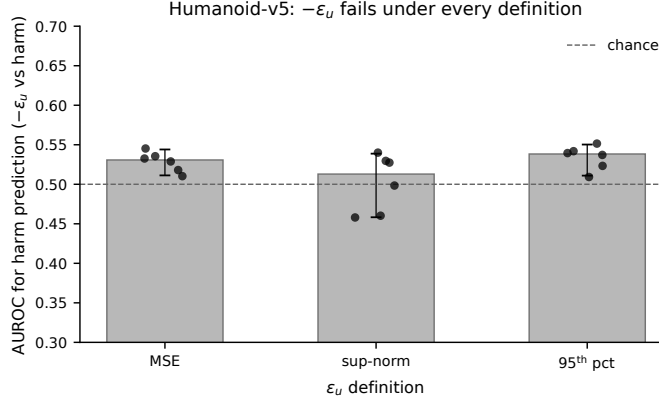


Figure 6: $-\hat{\epsilon}_u$ fails as a harm predictor under three definitions on Humanoid-v5. Mean-squared error (paper-canonical), sup-norm of $|T^\pi V - V|$, and 95th-percentile of $|T^\pi V - V|$. Six seeds; all three definitions sit at or near chance.

380 E Robustness of ϵ_u across scalarizations

381 The headline ϵ_u in Table 1 uses mean-squared TD error post-critic-update (the value-function loss
 382 every PPO and SAC implementation we surveyed minimizes). Figure 6 reports the same quantity
 383 computed under two alternative scalarizations on Humanoid-v5: the sup-norm $\|T^\pi V - V\|_\infty$
 384 (corresponding to the worst-case theoretical ϵ_u) and the 95th-percentile of $|T^\pi V - V|$ (a robust
 385 variant). All three sit at chance. The failure of $-\hat{\epsilon}_u$ as a per-update harm predictor is intrinsic to the
 386 metric, not to the choice of scalarization.

387 F Per-seed paired comparison

388 Table 1 in the main text reports per-(algorithm, environment) median AUROC. Figure 7 visualizes
 389 the seed-paired comparison for the PPO main grid.

390 G Off-policy advantage estimation via resampling

391 To compute $\hat{A}_k$ for off-policy SAC without importance weighting, we use the continuous-action
 392 resampling bypass. We define the soft value function under the pre-update critic Q_k as:

$$V_k^{\text{soft}}(\pi, s) := \mathbb{E}_{a \sim \pi} [Q_k(s, a) - \alpha \log \pi(a|s)]. \quad (7)$$

393 Given a replay buffer $\mathcal{D}$, our estimator is:

$$\hat{A}_k^{\text{resampled}} := \mathbb{E}_{s \sim \mathcal{D}} [V_k^{\text{soft}}(\pi_{k+1}, s) - V_k^{\text{soft}}(\pi_k, s)]. \quad (8)$$

394 By sampling actions from the current policy rather than replaying behavior actions, we eliminate the
 395 π_{k+1}/β importance weights and their associated variance explosion. However, this is an approxi-
 396 mation: it evaluates the new policy π_{k+1} under the old critic Q_k rather than the post-update critic
 397 Q_{k+1} . While it eliminates the importance-weight variance, it introduces a bias whose direction and
 398 magnitude we do not fully characterize, and serves as a practical diagnostic rather than a perfect
 399 population statistic.

400 H Generalized Advantage Estimation and the λ confound

401 One might ask if the distribution mismatch is modulated or masked by Generalized Advantage
 402 Estimation [Schulman et al., 2015b]. GAE uses a parameter $\lambda \in [0, 1]$ to interpolate between
 403 one-step TD errors ($\lambda = 0$) and Monte Carlo returns ($\lambda = 1$). While lower λ effectively mixes
 404 distributions by introducing bias toward the current critic, our audit uses the standard $\lambda = 0.95$ across

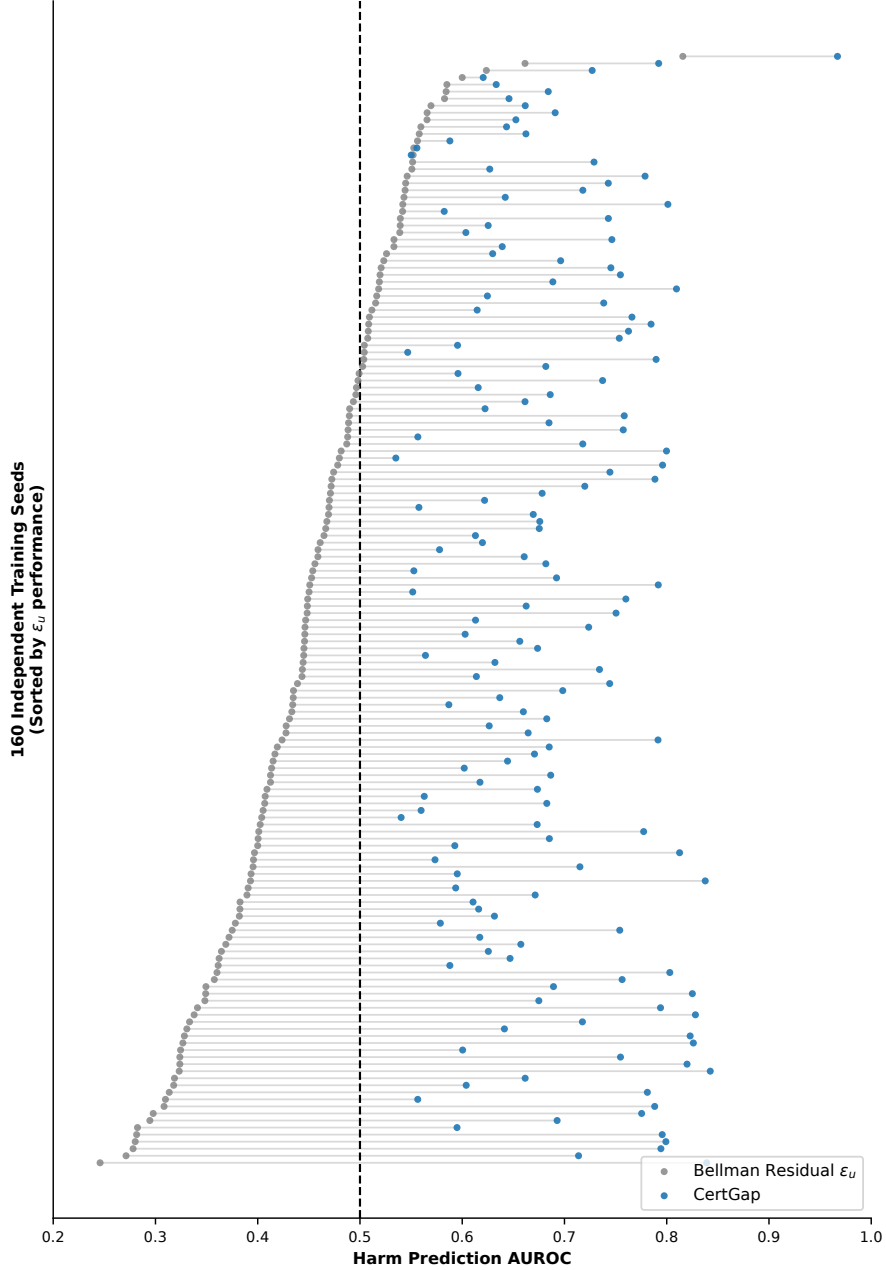


Figure 7: **Per-seed pairwise comparison.** Paired dumbbell plot for the 160 PPO seeds (sorted by $\hat{\epsilon}_u$ AUROC). Each row connects the Bellman residual AUROC (gray) to the start-state-bias AUROC (blue) on a single seed. The bias term beats the residual on 158/158 paired seeds (one tied), $p < 10^{-27}$.

all PPO and TRPO environments. The structural argument implies that the d^π vs. ν gap remains the dominant source of monitor failure regardless of the λ setting: the structural mismatch at ν is not a property of the advantage estimator but of the policy improvement objective itself.

I Prospective limits of rollout-only diagnostics

Theorem 1 (Rollout-only diagnostics cannot prospectively certify improvement). *Let $\mathcal{A}$ be any actor-critic update algorithm that maps a policy and a finite rollout batch to a new policy, $\pi_{k+1} =$*

411 $\mathcal{A}(\pi_k, B_k)$, and relies on a critic $V_{k+1} = \arg \min_V \mathcal{L}(V; B_k)$. Let $\phi_k = \phi(V_{k+1}, B_k, \pi_k, \pi_{k+1})$ be
 412 any scalar diagnostic available at the conclusion of step k . Assume:

413 • **Exploration:** The rollout $B_k \sim \pi_k$ leaves at least one state s_{new} unvisited with probability
 414 1, but the updated policy π_{k+1} visits s_{new} with positive probability.

415 • **Fallibility:** $\mathcal{A}$ is not a constant map and is susceptible to either finite-sample variance or
 416 function-approximation interference (catastrophic forgetting).

417 Then there exist two MDPs M_+ and M_- such that $\phi_k(M_+) = \phi_k(M_-)$ exactly, but $\mathbb{E}[\Delta J_{k+1} \mid$
 418 $M_+] > 0$ while $\mathbb{E}[\Delta J_{k+1} \mid M_-] < 0$.

419 **Proof. Step 1 — Identical Rollouts and Diagnostics.**

420 Let $\mathcal{S}_k = \text{supp}(d^{\pi_k})$ be the support of π_k 's state visitation measure. Construct M^+ and M^- to agree
 421 with a common base MDP M_0 on all $s \in \mathcal{S}_k$ (same transition kernel and reward function). By the
 422 Exploration assumption, $s_{\text{new}} \notin \mathcal{S}_k$ with probability 1, so every trajectory in $B_k \sim \pi_k$ is confined to
 423 $\mathcal{S}_k$:

$$P(B_k \mid M^+, \pi_k) = P(B_k \mid M^-, \pi_k)$$

424 Since $V_{k+1} = \arg \min_V \mathcal{L}(V; B_k)$ and $\pi_{k+1} = \mathcal{A}(\pi_k, B_k)$ are deterministic functions of (B_k, π_k)
 425 alone, they are identical in both MDPs. Therefore:

$$\phi_k(M^+) = \phi(V_{k+1}, B_k, \pi_k, \pi_{k+1}) = \phi_k(M^-) \quad (\star)$$

426 **Step 2 — Constructing M^+ with $\mathbb{E}[\Delta J_{k+1} \mid M^+] > 0$.**

427 Choose M_0 so that no action in M_0 transitions deterministically to s_{new} from any $s \in \mathcal{S}_k$. In M^+ ,
 428 configure s_{new} with reward $R > 0$ transitioning to an absorbing zero-reward state. By Exploration,
 429 π_{k+1} visits s_{new} with positive probability, and there exists a predecessor $s_{\text{prev}} \in \mathcal{S}_k$ from which π_{k+1}
 430 takes action $a \rightarrow s_{\text{new}}$ with probability $q := \pi_{k+1}(a \rightarrow s_{\text{new}} \mid s_{\text{prev}}) \in (0, 1)$.

431 The advantage of $a \rightarrow s_{\text{new}}$ at s_{prev} is:

$$A_{M^+}^{\pi_{k+1}}(s_{\text{prev}}, a \rightarrow s_{\text{new}}) = R - V_{M^+}^{\pi_{k+1}}(s_{\text{prev}})$$

432 Since $V_{M^+}^{\pi_{k+1}}(s_{\text{prev}}) = q \cdot R + (1 - q) \cdot C_{\text{avg}}$ where $C_{\text{avg}} \leq C_{M_0} := \max_{s,a} r_{M_0}(s, a)/(1 - \gamma) < \infty$,
 433 this simplifies to $(1 - q)(R - C_{\text{avg}})$, which is strictly positive for any $R > C_{M_0}$ (a free parameter
 434 choice).

435 Since π_{k+1} was computed from B_k containing no s_{new} transitions, it is suboptimal for M^+ . At step
 436 $k + 1$, $B_{k+1} \sim \pi_{k+1}$ reveals s_{new} 's reward R . For R large enough that the advantage signal at s_{prev}
 437 dominates finite-sample noise, algorithm $\mathcal{A}$ (which is not a constant map and uses the advantage
 438 signal to update — the defining property of actor-critic methods) shifts π_{k+2} toward $a \rightarrow s_{\text{new}}$ at s_{prev}
 439 by some $\delta > 0$. By the Performance Difference Lemma:

$$J(\pi_{k+2} \mid M^+) - J(\pi_{k+1} \mid M^+) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^{\pi_{k+2}}} [A_{M^+}^{\pi_{k+1}}(s, \pi_{k+2})]$$

440 The contribution at s_{prev} is $d^{\pi_{k+2}}(s_{\text{prev}}) \cdot (1 - q)(R - C_{\text{avg}}) \cdot f(\delta) > 0$ where $f(\delta) > 0$ for any $\delta > 0$.
 441 Contributions from other $s \in \mathcal{S}_k$ are $O(C_{M_0})$ in magnitude. For R sufficiently large, the s_{prev} term
 442 dominates, giving:

$$\mathbb{E}[J(\pi_{k+2} \mid M^+) - J(\pi_{k+1} \mid M^+)] > 0$$

443 **Step 3 — Constructing M^- with $\mathbb{E}[\Delta J_{k+1} \mid M^-] < 0$.**

444 The Fallibility assumption guarantees $\mathcal{A}$ fails under at least one of the following two mechanisms. In
 445 each sub-case, M^- is constructed to agree with M^+ on $\mathcal{S}_k$, preserving $(\star)$, while configuring s_{new} to
 446 trigger the failure.

447 **Sub-case A: Function-Approximation Interference.**

448 By Fallibility (susceptibility to catastrophic forgetting), there exist an MDP $\widetilde{M}$, a policy $\tilde{\pi}$ with
 449 $\text{supp}(d^{\tilde{\pi}}) =: \mathcal{S}_k$, a state $\tilde{s} \notin \mathcal{S}_k$ reachable under $\tilde{\pi}' = \mathcal{A}(\tilde{\pi}, \widetilde{B})$ (consistent with Exploration), and a
 450 reward value $\tilde{r}$ at $\tilde{s}$, such that:

$$\mathbb{E}_{\widetilde{B}, \widetilde{B}'} [J(\mathcal{A}(\tilde{\pi}', \widetilde{B}')) - J(\tilde{\pi}')] < 0$$

451 The mechanism: the gradient $\nabla_{\phi} \mathcal{L}(V_{\phi}; \tilde{\mathcal{B}}')$, which includes transitions at $\tilde{s}$ with reward $\tilde{r}$, is anti-
 452 aligned with the gradient from critical states $s_i \in \mathcal{S}_k$ due to shared function-approximator parameters.
 453 The joint gradient step degrades V^* estimates at s_i , causing π_{k+2} to take suboptimal actions there.

454 Set $\pi_k := \tilde{\pi}$, $M_0 := \tilde{M}|_{\mathcal{S}_k}$, $s_{\text{new}} := \tilde{s}$, $r_{M^-}(s_{\text{new}}) := \tilde{r}$, and $M^- := \tilde{M}$. By construction, M^-
 455 agrees with M^+ on $\mathcal{S}_k$ (both equal $\tilde{M}|_{\mathcal{S}_k}$), so $(\star)$ is preserved. The failure guaranteed in $\tilde{M}$ applies
 456 directly to M^- :

$$\mathbb{E}[J(\pi_{k+2} | M^-) - J(\pi_{k+1} | M^-)] < 0$$

457 Sub-case B: Finite-Sample Variance.

458 By Fallibility (susceptibility to finite-sample variance), there exist $\tilde{M}$, $\tilde{\pi}$ with $\text{supp}(d\tilde{\pi}) =: \mathcal{S}_k$,
 459 $\tilde{s} \notin \mathcal{S}_k$ reachable under $\tilde{\pi}' = \mathcal{A}(\tilde{\pi}, \tilde{\mathcal{B}})$, and reward distribution ν with $\mathbb{E}_{\nu}[r] \geq 0$ and $\text{Var}_{\nu}[r] \gg 0$,
 460 such that:

$$\mathbb{E}_{\tilde{\mathcal{B}}, \tilde{\mathcal{B}}'}[J(\mathcal{A}(\tilde{\pi}', \tilde{\mathcal{B}}')) - J(\tilde{\pi}')] < 0$$

461 The mechanism: the high-variance return signal from $\tilde{s}$ in $\tilde{\mathcal{B}}'$ overwhelms the correct policy gra-
 462 dient signal from $\mathcal{S}_k$ states (since $\mathcal{A}$ is not constant, it responds to this signal), pushing π_{k+2} in a
 463 performance-degrading direction at states in $\mathcal{S}_k$.

464 Set $\pi_k := \tilde{\pi}$, $M_0 := \tilde{M}|_{\mathcal{S}_k}$, $s_{\text{new}} := \tilde{s}$, $r_{M^-}(s_{\text{new}}) \sim \nu$, and $M^- := \tilde{M}$. Agreement on $\mathcal{S}_k$ preserves
 465 $(\star)$, and:

$$\mathbb{E}[J(\pi_{k+2} | M^-) - J(\pi_{k+1} | M^-)] < 0$$

466 Conclusion.

467 In both sub-cases: $(\star)$ yields $\phi_k(M^+) = \phi_k(M^-)$, while Steps 2 and 3 yield $\mathbb{E}[\Delta J_{k+1} | M^+] > 0$
 468 and $\mathbb{E}[\Delta J_{k+1} | M^-] < 0$. Since the diagnostic cannot distinguish M^+ from M^- , yet the sign of
 469 $\mathbb{E}[\Delta J_{k+1}]$ differs between them, no scalar $\phi_k = \phi(V_{k+1}, \mathcal{B}_k, \pi_k, \pi_{k+1})$ can prospectively certify
 470 that $\mathbb{E}[\Delta J_{k+1}] > 0$ across all MDPs satisfying the theorem’s conditions. $\square$

471 J Experimental details

472 **Environments and seed counts.** PPO main: LunarLander-v3, CartPole-v1, Acrobot-v1 at 200k–
 473 300k env steps; Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5 at 300k steps; Humanoid-v5 at
 474 500k steps; 20 seeds per environment (160 runs). PPO-MSE ablation: same eight envs, 5 seeds each
 475 (40 runs). Hyperparameter factorials (PPO): LunarLander 4×4 , 5 seeds/cell (80 runs); HalfCheetah
 476 3×2 plus 3 value-lr cells, 5 seeds each (45 runs); Hopper 3×2 plus 3 value-lr cells, 5 seeds each (45
 477 runs). ϵ_u -scalarization subset (Humanoid): 6 seeds with three Bellman-residual scalarizations logged.
 478 Estimator-agreement subset (Humanoid): 20 seeds with 500-transition held-out batch logging. SAC:
 479 Hopper-v5, HalfCheetah-v5, Walker2d-v5, Ant-v5, Humanoid-v5, 6–10 seeds each (45 runs). TRPO:
 480 LunarLander-v3, Hopper-v5, 10 seeds each (20 runs). Cross-algorithm total: ~ 560 training runs,
 481 $\sim 90,000$ logged updates. Environments via Gymnasium [Towers et al., 2024].

482 **Algorithm configurations.** PPO: GAE [Schulman et al., 2015b] with $\gamma=0.99$, $\lambda=0.95$; value-
 483 epochs 10, policy-epochs 10, clip- $\epsilon = 0.2$; batch size 2048; minibatch 64; policy learning rate
 484 3×10^{-4} , value learning rate 10^{-3} ; gradient clipping at 1.0; no entropy bonus. (64, 64)-tanh MLPs
 485 with orthogonal initialization. Continuous-action environments use a Gaussian policy with state-
 486 independent log-standard-deviation. PPO-MSE replaces the clipped value loss with raw MSE;
 487 otherwise identical. PPO-NU adds a start-state MSE loss term with coefficient 1.0, sampled from a
 488 separate start-state buffer. SAC: standard twin-Q networks with target smoothing, automatic entropy
 489 temperature, replay buffer 10^6 , batch size 256, learning rate 3×10^{-4} , $\gamma = 0.99$, $\tau = 0.005$. TRPO:
 490 KL constraint $\delta = 0.01$, conjugate-gradient 10 iterations, line-search backtracks 10, policy step
 491 from natural gradient; same value network and GAE configuration as PPO. All implementations in
 492 PyTorch on CPU.

493 **Diagnostic logging.** Per update we log $\hat{A}_k$, $\hat{\Delta}_k$ (start-state form), $\hat{\epsilon}_u$ (mean squared TD error on
 494 rollout transitions, post-critic-update), and $\hat{\Delta J}_k = \hat{J}(\pi_{k+1}) - \hat{J}(\pi_k)$ from successive rollout batches.

495 The harmful label $1[\widehat{\Delta J}_k < 0]$ is undefined on the final update of each run. The same logging schema
496 applies to PPO, PPO-MSE, SAC, and TRPO with appropriate definitions of “rollout batch” and the
497 post-update critic.

498 **Reproducibility.** A single command (`make figures` after `pip install -e .`) regenerates
499 every figure in this paper from the released update logs. The full pipeline (`make experiments`,
500 equivalently `python scripts/run_all.py -workers 6 -skip-if-exists`) takes ~ 13 wall-
501 clock hours (summed over the parallel-worker schedule) on an 8-core CPU (6 parallel workers due to
502 memory/IO). Code, run logs, and `REPRODUCIBILITY.md` are released with the paper.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Section 1 state the three contributions (no-go for absolute residuals; identity exposing $\hat{\Delta}_k$; cross-algorithm audit) with the empirical headline numbers; Sections 5 and 6 deliver each claim with the exact scope described.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 8 (“Limitations” paragraph) discusses the high variance of $\hat{J}(\pi_k)$ on environments with few episodes per rollout, the diagnostic-versus-controller boundary (lag-1 collapse), the smaller SAC/TRPO grids, and the absence of off-policy importance correction for A_k .

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: Proposition 1 is proved in the body. Proposition 2 is proved in Section C. Assumptions on the MDP and policy class are stated in Section 3.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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Justification: Section J specifies environments, seed counts, hyperparameters, and the per-update logging schema. The released code reproduces every figure and table with a single command.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: An anonymized code archive accompanies the submission. `README.md` documents both the figures-from-cache pipeline (`make figures`) and the from-scratch pipeline (`make experiments`).

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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Justification: Section J lists all hyperparameters; Section 4 describes the logging schema and harmful-label definition.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: Per-seed AUROCs are reported with paired-Wilcoxon p -values across 263 seeds (Table 1); the dominance counts and pooled p -values are reported in Section 6.

552 **8. Experiments compute resources**

553 Question: For each experiment, does the paper provide sufficient information on the com-
554 puter resources (type of compute workers, memory, time of execution) needed to reproduce
555 the experiments?

556 Answer: [Yes]

557 Justification: Section J reports CPU-only execution and a ~ 13 wall-clock-hour budget on
558 an 8-core CPU at six parallel workers.

559 **9. Code of ethics**

560 Question: Does the research conducted in the paper conform, in every respect, with the
561 NeurIPS Code of Ethics?

562 Answer: [Yes]

563 Justification: The work is purely methodological RL on synthetic and standard Gymnasium
564 environments; there is no human-subjects data, deployed model, or dual-use concern.

565 **10. Broader impacts**

566 Question: Does the paper discuss both potential positive societal impacts and negative
567 societal impacts of the work performed?

568 Answer: [N/A]

569 Justification: Improving training stability of RL has indirect implications for downstream
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572 Question: Does the paper describe safeguards that have been put in place for responsible
573 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
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575 Answer: [N/A]

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582 Answer: [Yes]

583 Justification: Gymnasium and MuJoCo are cited (Section J); PPO, SAC, TRPO, and GAE
584 are cited where they appear. All third-party software is MIT/BSD-compatible with our MIT
585 release.

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